# REB, The Book
# Example 6.6.4 Solution

An adiabatic, steady-state CSTR with a volume of 500 $cm^3$ is going to be used to convert A and B into Y and Z according to reaction (1). A liquid solution flowing at 1.0 $cm^3\ s^{-1}$ and containing equal amounts of A and B (0.015 mol $cm^{-3}$) at 50 °C will be used. The heat capacity of the fluid is essentially equal to that of the solvent, and can be considered to be constant and equal to 0.35 cal $g^{-1}\ K^{-1}$. The density of the fluid, also constant, is 0.93 g $cm^{-3}$. The rate expression for reaction (1) is given in equation (2), and the heat of reaction (1) may be assumed to be constant and equal to –20 kJ $mol^{-1}$. The rate coefficient displays Arrhenius temperature dependence with a pre-exponential factor equal to 3.24 x $10^{12}$ $cm^3\ mol^{-1}\ s^{-1}$, and an activation energy of 105.0 kJ $mol^{-1}$. How many steady states are possible at these conditions, and what are the conversion and outlet temperature for each of them?

$$A + B \rightarrow Y + Z \qquad (1)$$

$$r_1 = k_1 C_A C_B \qquad (2)$$

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***Assignment Summary***

Reaction

$$A + B \rightarrow Y + Z \qquad (1)$$

Rate Expression

$$r_1 = k_1 C_A C_B \qquad (2)$$

Reactor: Adiabatic, steady-state CSTR

Given Constants: $V$ = 500 $cm^3$, $\dot{V}_{in}$ = 1.0 $cm^3\ s^{-1}$, $C_{A,in}$ = 0.015 mol $cm^{-3}$, $C_{B,in}$ = 0.015 mol $cm^{-3}$, $T_{in}$ = (50 + 273.15) K, $\tilde{C}_p$ = 0.35 cal $g^{-1}\ K^{-1}$, $\rho$ = 0.93 gm $cm^{-3}$, $\Delta H_1$ = –20 kJ $mol^{-1}$, $k_{0,1}$ = 3.24 x $10^{12}$ $cm^3\ mol^{-1}\ s^{-1}$, and $E_1$ = 105.0 kJ $mol^{-1}$.

Parameter:

$T$ for multiplicity plot

Deliverables: $f_A$ and $T$ for each of 3 steady states

### *Completion Strategy*

This assignment involves two distinct response tasks that will be completed sequentially. The first is the generation of a multiplicity plot, and the second is the calculation of the conversion and outlet temperature at each of three steady states. To generate the multiplicity plot, the design equations will be solved to find the outlet molar flow rates and inlet temperature for a range of outlet temperatures. Plotting the results will yield a multiplicity plot. To analyze the three steady states, the design equations will be solved for the outlet molar flow rates and outlet temperature. Different initial guesses for the outlet temperature will be required to find the three solutions.

### *CSTR Model Functions*

CSTR Design Equations:

$$0 = \dot{n}_{A,in} - \dot{n}_A - Vr_1 = \epsilon_1 \tag{3}$$

$$0 = \dot{n}_{B,in} - \dot{n}_B - Vr_1 = \epsilon_2 \tag{4}$$

$$0 = -\dot{n}_Y + Vr_1 = \epsilon_3 \tag{5}$$

$$0 = -\dot{n}_Z + Vr_1 = \epsilon_4 \tag{6}$$

$$0 = -\dot{V}_{in}\rho\tilde{C}_p\left(T - T_{in}\right) - Vr_1\Delta H_1 = \epsilon_5 \tag{7}$$

CSTR Model Variables:

$\dot{n}_A$, $\dot{n}_B$, $\dot{n}_Y$, $\dot{n}_Z$, and $T_{in}$ for multiplicity plot

$\dot{n}_A$, $\dot{n}_B$, $\dot{n}_Y$, $\dot{n}_Z$, and $T$ for calculating steady states

Additional Computable Unknowns: $\dot{n}_{A,in}$, $r_1$, $k_1$, $C_A$, $\dot{V}$, $C_B$, and $\dot{n}_{B,in}$

$$\dot{n}_{i,in} = \dot{V}_{in}C_{i,in} \qquad i = A \text{ and } B \tag{8}$$

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT}\right) \tag{9}$$

$$\dot{V} = \dot{V}_{in} \tag{10}$$

$$C_i = \frac{\dot{n}_i}{\dot{V}} \qquad i = A \text{ and } B \tag{11}$$

Additional Uncomputable Unknown: $T$ for multiplicity plot

ATE Solver Requirements:

1. Initial guesses for the CSTR model variables (provided as an argument to the CSTR model function)
2. $T$ must be made available to the residuals function for multiplicity plot
3. Residuals function that
   a. receives guesses for the reactor model variables ($\dot{n}_A$, $\dot{n}_B$, $\dot{n}_Y$, $\dot{n}_Z$, and $T_{in}$ for the multiplicity plot and $\dot{n}_A$, $\dot{n}_B$, $\dot{n}_Y$, $\dot{n}_Z$, and $T$ for the steady state responses)
   b. calculates the additional unknowns using equations (2), (8) - (10)
   c. evaluates and returns the CSTR design equation residuals, equations (3) - (7)

## ***Deliverables Functions***

1. For the multiplicity plot
   a. Define a range of values for $T$

$$\underline{T} = [-25, \cdots, 350] + 273.15K \tag{14}$$

   b. Define an initial guess for the CSTR model variables for the first value of $T$

$$\dot{n}_{i,guess} = \dot{n}_{i,in} \qquad i = A, B, Y, \text{ and } Z \tag{15}$$

$$T_{in,guess} = T - 1K \tag{16}$$

   c. Loop through the values of $T$ and
      i. Call the CSTR model function, passing the initial guess as an argument, to solve the design equations for the CSTR model variables
      ii. Saves the value of $T_{in}$

iii. Sets the initial guess equal to the current solution of the design equations

d. P Plot $\underline{T}$ *vs.* $\underline{T}_{in}$

2. For calculating steady states

a. Define initial guesses for the three steady states

$$\dot{n}_{i,guess} = \dot{n}_{i,in} \qquad i = A, B, Y, \text{ and } Z \tag{17}$$

$T_{guess,low}$ read from multiplicity plot (18)

$T_{guess,mid}$ read from multiplicity plot (19)

$T_{guess,high}$ read from multiplicity plot (20)

b. For each steady state

i. Call the CSTR model function, passing the initial guess as an argument, to solve the design equations for the CSTR model variables

ii. Calculate the conversion

$$f_{A,low} = \frac{\dot{n}_{A,in} - \dot{n}_{A,low}}{\dot{n}_{A,in}} \tag{21}$$

$$f_{A,mid} = \frac{\dot{n}_{A,in} - \dot{n}_{A,mid}}{\dot{n}_{A,in}} \tag{22}$$

$$f_{A,high} = \frac{\dot{n}_{A,in} - \dot{n}_{A,high}}{\dot{n}_{A,in}} \tag{23}$$

### *Implementation of the Calculations*

The Python scripts, example_6_6_4_part_A.py and example_6_6_4_part_B.py, and Matlab functions, example_6_6_4_part_A.m and example_6_6_4_part_B.m, that were written to perform the calculations for the multiplicity and steady states, respectively, accompany this solution.

## *Results and Discussion*

When generating the multiplicity plot, the ATE solver converged with the initial guess in equation (15), and the range of values chosen in equation (14) spanned an appropriate range for the multiplicity plot, which is shown in Figure 1where it can be seen that there indeed are three steady states for inlet temperatures between ca. 4 and 90 °C. The solid red line represents an ignition point and the dashed red line, an extinction point. It is expected that the middle steady states, shown as a dashed black line, will be unstable. Guesses for the steady-state outlet temperatures at an inlet temperature of 50 °C (50, 140, and 270 °C) were read from the multiplicity plot as indicated in equations (17) through (19).

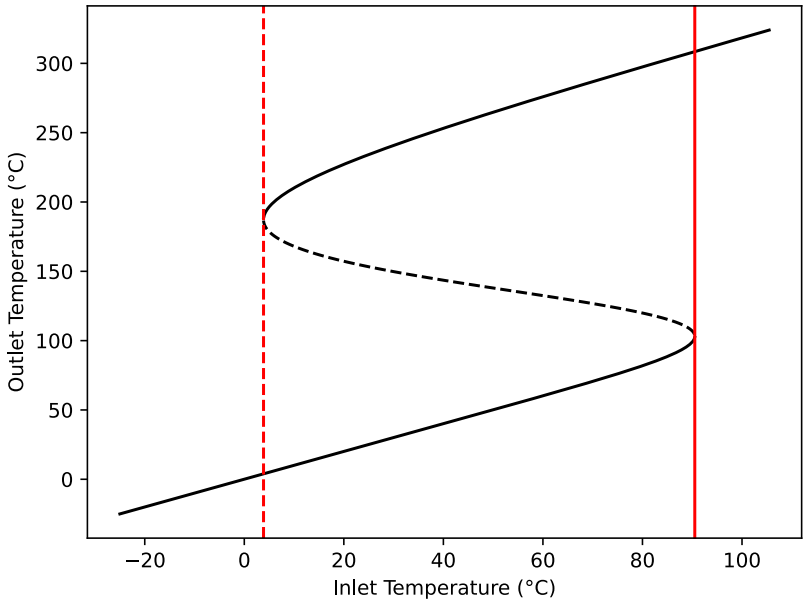



*Figure 1. Outlet temperature as a function of inlet temperature.*

When calculating the steady states, the ATE solver converged using the low temperature and middle temperature guesses (50 and 140 °C) read from the multiplicity plot. When the high temperature guess read from the multiplicity plot (270 °C) was used, the solver yielded a converged solution, but the conversion was over 100%.

Reducing the guess to 240 °C yielded a conversion less than 100%. The results are shown in Table 1.

Table 1. Steady-state conversions and outlet temperatures

| Conversion (%) | Outlet Temperature (°C) |
|---|---|
| 0.03 | 50.1 |
| 39.9 | 138 |
| 97.5 | 265 |


***Summary***

Three steady states are possible. There is a stable steady state with an outlet temperature of 50.1 °C where the conversion is 0.03%, and one at 265 °C with a conversion of 97.5%. A third, unstable steady state has an outlet temperature of 138 °C and a conversion of 39.9%.